MIL-E-1/1622A(USAF)
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1 SEPTEMBER 1966

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 7829A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex, integral cavity, 11.0 GHz, adjustable frequency, 60 mW, waveguide output, convection air cooled

ABSOLUTE RATINGS:

Parameter:	E_f	E_{rs}	E_r	T_{body}	Alt.
Unit:	V	Vdc	Vdc	°C	Ft
Maximum:	7.0	350	-500	200	10,000
Minimum:	5.7	---	Note 1	---	----

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1
Mounting position:	Any
Output flange:	Mates with UG-39/U and UG-40/U
Cathode:	Unipotential

TEST CONDITIONS:

Parameter:	Ef	Ers	Er	F
Unit:	V	Vdc	Vdc	MHz
	6.3	300 ±3	-200 to 400	11,000 ±5 Note 4

REQUIREMENT OR TEST: General

Preproduction Test - Required (See note 13)

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1					
4214	† Cathode emission	E_f = 5.7 V tk = 120 sec	$\Delta Ik/Ik$	---	15	%
1256	Resonator current	E_r/max Po (Note 1)	I_{rs}	---	35	mAdc
4213	Reflector voltage	Power output (Note 11)	E_r	-270	-340	Vdc
4223	Mechanical tuning range	E_r/max Po, Tuner turns = 2.0 turns Low frequency High frequency	 F F	 --- 11,050	 10,950 ---	 MHz MHz
4250	† Power output	E_r/max Po; E_r = -270 to -340 Vdc (Notes 2 and 9)	Po	60	---	mW
4226	Warmup time	Po = 60 mW (Note 3)	t	10	15	min.
4229	† Total Reflector current	(Notes 8 and 10)	I_r	---	5	μAdc
	Quality conformance inspection, part 2					
1301	Heater current	E_f = 6.3 V	I_f	350	500	mAdc
4027	Temperature coefficient	E_r/max Po, TA = -10° +40°C (Note 6)	$\Delta F/\Delta C$	---	±200	kHz/°C
	Quality conformance inspection, part 3					
----	Life test	Group D, Power output conditions (Note 9)	t	500	---	hrs
----	Life test end point:					
4250	Power output	E_r/max Po	Po	48	---	mW
	Periodic Check Tests					
1027	Temperature cycling	Nonoperating, TA = -29° to +60°C (Note 5)	Cycles	5	---	---
1031	Variable frequency vibration	G = 2.5, F = 20 to 50 Hz (Note 7)	---	---	---	----

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NOTES:

1. The reflector voltage must always be negative with respect to the cathode by at least 20 volts.

2. All oscillation tests, except vibration, shall be made with the tube rigidly connected to a UG-39/U flange on appropriate RG-52/U waveguide equipment and the load VSWR for the test shall be less than 1.1:1. All oscillation tests shall be under test conditions except where noted.

3. With the tube body temperature stabilized at room temperature, 25° ±5°C, and the reflector voltage set to the value obtained during performance of Method 4223, mechanical tuning, the tube shall meet specified conditions within the specified limits of warmup time.

4. The frequency shall be adjusted and measured under test conditions. The tube under test shall be stabilized for 10 minutes at:

   Heater voltage = 6.3 ±0.2 V

   Resonator voltage = 300 ±3 Vdc

   Ambient temperature = 25° ±5°C

5. Tubes under test shall meet all specification tests after this test.

6. The tube under test shall be operated at initial frequency, F = 11,000 ±5 MHz, under Method 4250 conditions. The initial frequency shall be allowed to stabilize at practically constant ambient temperatures for at least 10 minutes. The rate of change of ambient temperature shall not exceed 2°C per minute.

7. The tube under test shall be nonoperating and rigidly mounted on the vibration table. The frequency range, 20 to 55 Hz, shall be traversed in approximately 1 minute. The motion shall be applied for a period of 30 minutes in each of the three mutually perpendicular planes, one of which shall be parallel to the axis of the beam.

8. The tube under test shall not be oscillating during this test.

9. Forced air cooling shall not be used.

10. Apply heater voltage for 2 minutes. Then apply other voltages plus heater voltage for 2 minutes and read the total reflector current at the end of the latter 2-minute period.

11. The frequency shall be allowed to stabilize by operation at ambient temperature, 25° ±5°C, for at least 10 minutes. After stabilization, the reflector voltage shall be adjusted for maximum power output. The frequency and reflector voltage shall be within the limits specified.

12. For quality conformance inspection part 1, AQL 0.65, inspection level II, one percent combined defective shall apply. For quality conformance inspection part 2, AQL 6.5, inspection level L6 shall apply.

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NOTES: Continued

13. The first article (preproduction) sample required for inspection and the acceptance criteria shall consist of four tubes. Life test shall subsequently be performed on three of the four sample tubes. First article testing shall include all tests and requirements of this specification and table 1 of MIL-E-1, where required. The four sample tubes shall be subjected to all electrical tests prior to undergoing environmental testing. The contractor/ manufacturer shall provide test reports and written assurance that:

    a. The first article sample is representative of the product to be submitted under the contract.

    b. Production articles will be manufactured from new materials and in the facility that produced the sample tubes.

    A test plan shall be submitted to the contracting officer for approval prior to proceeding with first article testing. The test plan shall list all production test and laboratory test facilities to be utilized in first article testing and production phases of the contract. Upon receipt of approval of the test plan, first article testing may proceed. When testing has been completed, three copies of the test report shall be prepared in accordance with MIL-STD-831, certified by a Government Representative and submitted with the test samples to the contracting officer for approval.

Custodian:

Air Force - 85

Review activity:

Air Force - 80

User Activity:

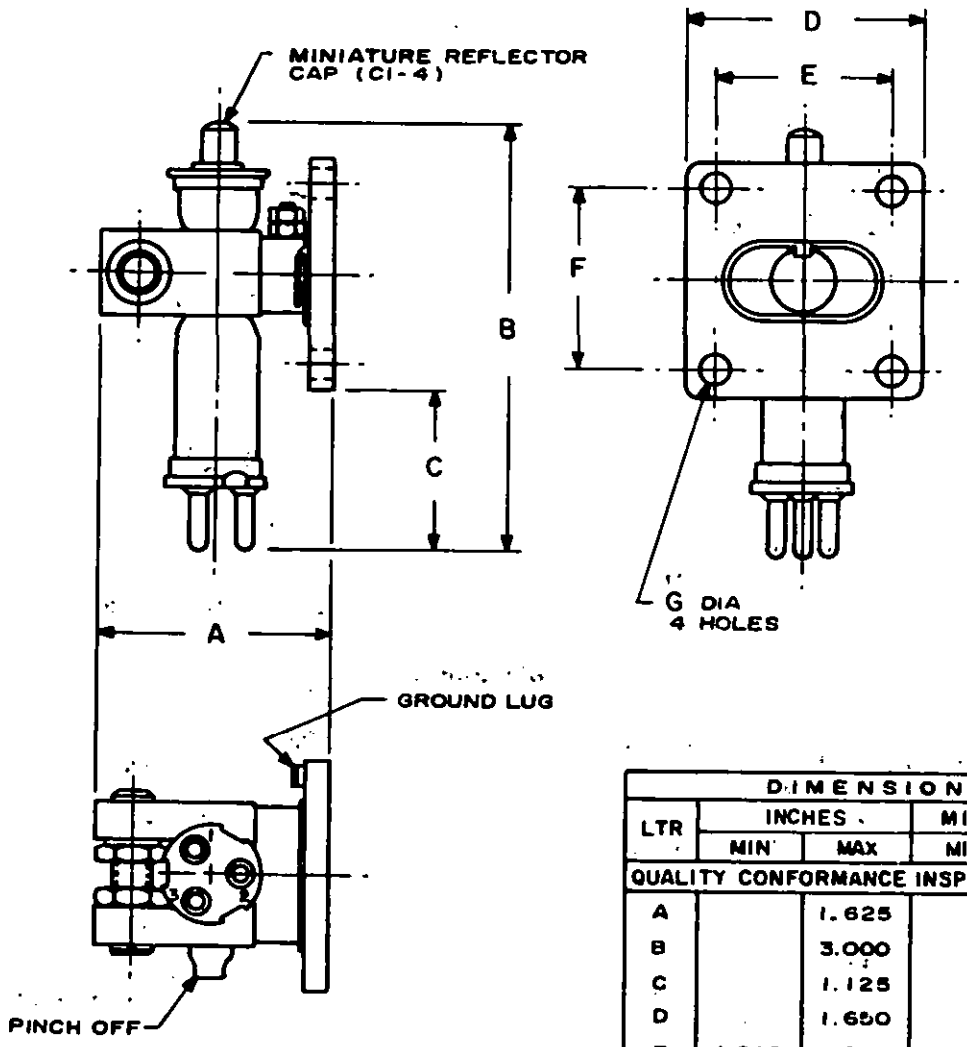
Air Force - 11

Preparing activity:

Air Force - 85

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DIMENSIONS				
LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSPECTION PART I				
A		1.625		41.28
B		3.000		76.20
C		1.125		28.58
D		1.650		41.91
E	1.216	1.224	30.89	31.09
F	1.276	1.284	32.41	32.61
G	.215	.225	5.46	5.72

PIN CONNECTIONS	
NO.	ELEMENT
o 1	HEATER
o 2	CATHODE
3	HEATER

o INTERNALLY CONNECTED

NOTES:

a. METRIC EQUIVALENTS (TO THE NEAREST .01 mm) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 mm.

b. JEDEC BASE TYPE PEE WEE WAFER BASE A3-108 FOR DIMENSIONAL REFERENCES ONLY.

c. OUTPUT CONNECTOR MATES WITH UG-39/U OR UG-40A/U FLANGES.

FIGURE 1 Outline drawing of type 7829A

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-E-1/1622A (USAF) Electron Tube, Klystron Type 7829A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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